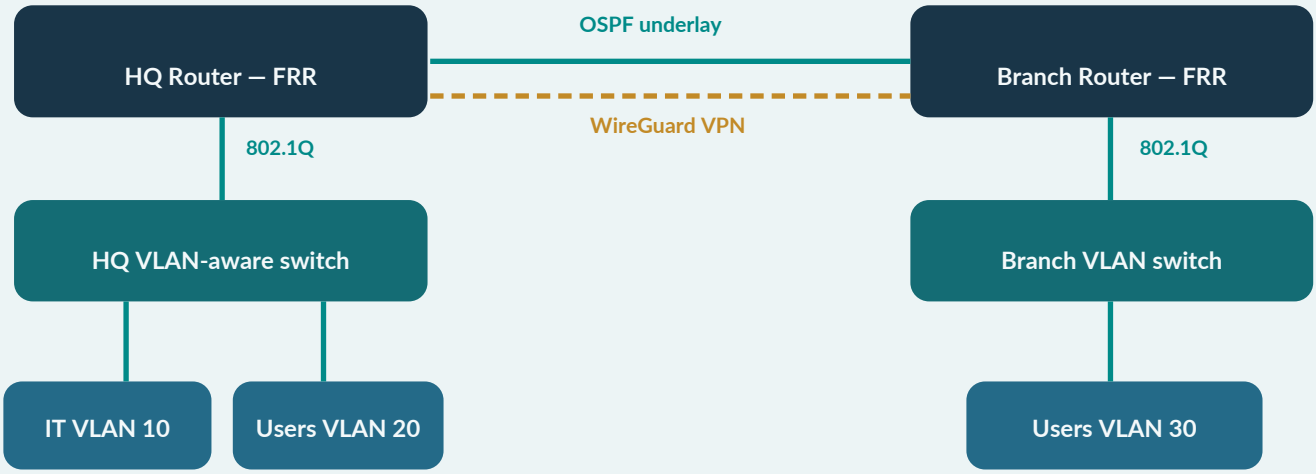
Enterprise Multi-Site Network Lab

V1 · Configuration package completed · Live validation pending

OBJECTIVE

Design an isolated enterprise topology joining a Headquarters (HQ) and Branch office. The lab source demonstrates dynamic routing, VLAN-based segmentation, VPN overlay routing and rule-based access control. It includes automated verification scripts and documentation, but it has not yet been executed on the target Linux VM.

NETWORK ARCHITECTURE



DESIGNED TECHNOLOGY STACK

Routing / L2

FRRouting, OSPF area 0, IPv4 subnets, 802.1Q trunks and Linux bridge access ports.

Security / Operations

WireGuard site-to-site overlay, iptables network restrictions, acceptance tests, fail-closed check.

Documentation and source code: github.com/zitanet44-commits (project repository publication pending)

Security controls & validation plan

V1 · Configuration package completed · Live validation pending

SECURITY POLICY MATRIX

HQ IT → Branch Users	**ALLOW**	Encrypted route via WireGuard wg0
HQ Users → Branch Users	**ALLOW**	Encrypted route via WireGuard wg0
Branch Users → HQ Users	**ALLOW**	Encrypted route via WireGuard wg0
HQ Users → HQ IT	**DENY**	HQ iptables FORWARD policy
Branch Users → HQ IT	**DENY**	HQ iptables FORWARD policy
Inter-site traffic during VPN outage	**DENY**	WAN egress block (fail-closed)

ACCEPTANCE TEST CHECKLIST

- Static source checks — 4 checks passed in authoring container
- OSPF FULL adjacency and exchanged prefixes — pending lab execution
- VLAN tags, gateways, and access memberships — pending lab execution
- Encrypted VPN route preference and handshake — pending lab execution
- Allowed connectivity / blocked connectivity — pending lab execution
- Tunnel-failure test with no plaintext fallback — pending lab execution

ENGINEERING NOTES

Security: VPN credentials are created at runtime and stored in gitignored files. The educational OSPF WAN control plane is not production-authenticated. Neither hardware firewall appliances nor a SIEM are part of the current lab profile.

Evidence: Real screenshots, route tables, VPN handshakes, packet captures and troubleshooting reports must be collected only after the target VM is deployed. This case study intentionally reports no invented test outcomes.

Next milestone: Run the lab in Ubuntu, fix environment-specific issues, pin tested images, add redacted evidence and update the GitHub/portfolio release from source-prepared to verified.

Taha Zitane • Systems & Networks • Chefchaouen, Morocco • zitanet44@gmail.com